

Lucas Lu

B.Eng. Candidate in Industrial Engineering | Class of 2029
12511133@mail.sustech.edu.cn | (+86) 157-2864-3180
lucas12511133.github.io | Shenzhen, China

EDUCATION

Southern University of Science and Technology (SUSTech)	Aug. 2025 - Present
<i>B.Eng. in Industrial Engineering, Zhicheng College</i>	<i>Shenzhen, China</i>
GPA: 3.91 / 4.0 Ranking: 1 / 11. Selected Coursework: Mathematical Analysis I-II (93, 96), Ordinary Differential Equations B (98), Foundation of Probability Theory (94), Linear Algebra (89), Introduction to C Programming (96), College Physics II (90).	

RESEARCH INTERESTS

Operations Research & Optimization; Emergency Response & Logistics; Machine Learning & Data Science; Mathematical Modeling.

RESEARCH EXPERIENCE

EMS Station Location Optimization	2026 - Present
<i>Project Lead 120 EMS Station Location Optimization</i>	
- Led a 6-member team in optimizing EMS station locations across Shenzhen; partitioned the city into 5,279 500m x 500m grids. - Built an XGBoost ETA model from 620K+ dispatch records and 360K Tencent Maps samples (MAE = 1.64 min, R2 = 0.863), achieving 90% coverage within 10 minutes. - Finalist, 8th China University Mechanical Engineering Innovation & Creativity Competition (2026). Tools: Python, OSMnx, XGBoost.	
AED Deployment at Gulongzhong	2026 - Present
<i>Project Participant AED Deployment at Gulongzhong</i>	
- Compared fixed AED placement with a dynamic human-vehicle-drone response scheme using road-network analysis. - Explored sustainable operating models, including potential insurance partnerships.	
Meal Optimization WeChat Mini Program	2026
<i>Project Developer Personalized Meal Recommendation</i>	
- Built a personalized meal recommendation mini program with Taro + React + TypeScript, Tencent Cloud Functions, and an optimization solver; supports meal composition, taste matching, nutrition constraints, and set-meal recommendations. - Implemented upper/lower bounds for calories, protein, carbohydrates, and fats, plus hard exclusion of disliked ingredients, dish-level replacement, and alternatives for infeasible plans. - Added carbohydrate-coverage alerts and diet suggestions; improved reliability with legacy solver compatibility, exception fallbacks, and clearer frontend feedback.	

LEADERSHIP & VOLUNTEER EXPERIENCE

Core Student Leader	Oct. 2025 - Present
<i>SUSTech IE Hunt Optimization challenge planning and team coordination</i>	
Co-led Seasons 1-2 of a campus-wide optimization challenge, integrating the Traveling Salesman Problem and Linear Programming into game mechanics; coordinated a 5-member team across logistics, feasibility analysis, and promotion.	
Student Representative	Oct. 2025
<i>Zhicheng College 10th Anniversary Ceremony Sole freshman keynote speaker</i>	
Volunteer	2026
<i>APEC Shenzhen 2026</i>	
Group Leader	2025 - Present
<i>Orange Light Volunteer Service Team Community research and accessibility advocacy</i>	
Completed 160+ hours of volunteer service. Led community research across Shenzhen, Foshan, and Guangzhou; partnered with the Shenzhen Association for the Blind on accessibility supervision and hosted Oxford University delegations at SUSTech.	

SELECTED HONORS & AWARDS

- Scholarships:** First-Class Outstanding Student Scholarship, SUSTech (2026).
- Competitions:** First Prize, Greater Bay Area Industrial Engineering Innovation Competition (2026); Second Prize, National College Student Mathematics Competition (Non-Math A, 2025).
- Honors:** APRU ULP Outstanding Student Ambassador (one of ten university-wide, 2026); Outstanding Individual, SUSTech Winter Social Practice and Winter Visit to Alma Mater (2026); Outstanding Camper, 4th Xiancheng Program (2025); Social Impact Award and Outstanding Volunteer Service Organization, Chengguang Volunteer Team (2025).

SKILLS & INTERESTS

- Programming Skills:** Python, MATLAB, R, C, LaTeX, AnyLogic, XGBoost, scikit-learn, NumPy, Pandas, Matplotlib, OSMnx.
- Languages:** Mandarin (Native), English (Proficient), Korean (Daily basics).